

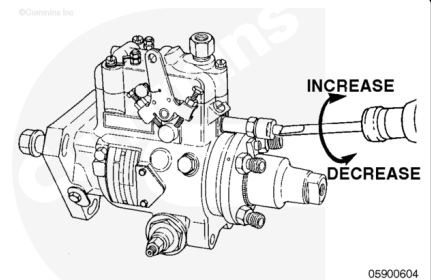
Trainings ▾

Adjust

Stanadyne

Stanadyne DB4 (Generator Application) Speed Droop Governor

Governor regulation of 3 percent to 5 percent can be attained with the speed droop governor. Precise control of governor regulation is done by decreasing or increasing the effective length of the governor control spring; this also decreases or increases the spring control spring rate. The governor control spring is threaded into an adjusting cap assembly and is referred to as the control rod assembly. By turning the adjusting cap in the **clockwise** direction, as viewed from the fuel transfer pump end, the control rod spring shortens and becomes less sensitive, thereby increasing governor regulation. Turning the adjusting cap in the **counterclockwise** direction increases the control rod spring length and sensitivity, which will decrease governor regulation.



The external speed droop adjustment screw, located at the rear of the fuel injection pump housing, controls the governor sensitivity. The droop screw adjustment varies the governor regulation by changing the effective spring rate. This adjustment will affect both full-load and no-load frequency settings and can require the high-speed stop screw be reset.

Speed droop adjustments **must** be made while the engine is operating. After each adjustment of the droop screw, the engine **must** be shutdown briefly in order to allow the governor spring to unload and the adjusting mechanism to seek its final position in the spring. Turning the screw in shortens the control spring, making it less sensitive and increasing speed droop. Turning the adjusting screw out has the opposite effect. Speed droop is the fuel injection pump's ability to respond to changing engine loads.

Speed Droop Adjustment

Note : If serious surging occurs during the warm-up period, turn the speed droop adjusting screw **clockwise**

until the surging stops.

Note : When the speed droop adjustments are made, it is necessary to adjust the throttle position.

Adjust the governor as follows:

1. Operate the engine until normal operating temperature is obtained 91°C [195°F].
2. When the engine reaches operating temperature, position the throttle to attain rated speed, and apply 100-percent load. Adjust the throttle position as necessary to obtain 100 percent performance.
3. Remove the load, and check for the specified no-load or, in the case of a generator set, note the frequency. If the no-load speed is incorrect, loosen the locking cap, and adjust the speed droop adjusting screw (**clockwise** for increased droop, **counterclockwise** for less droop). If surging occurs when the load is removed, turn the adjusting cap **clockwise** to eliminate the surge. Tighten the locking cap to secure the adjusting screw.
4. Check the 100-percent load and no-load performance again, and make adjustments as necessary.

